Ex No :6a Analysis of Prompting Techniques in LLMs Without Code

Zero-Shot Prompting

Q.No	Question	Prompt Type	Model Response	Actual Answer	Correct / Incorrect	Error Type	Remarks
1	A bakery has 60 loaves of bread. In the morning, 35% of the loaves are sold. How many loaves remain unsold?	Zero-shot	39 loaves	39 loaves	Correct	None	Single-step percentage problem; handled accurately without examples
2	Sam earns $18 per hour. Yesterday he worked for 6 hours. He spent $40 on groceries. How much money does Sam have left?	Zero-shot	$68	$68	Correct	None	Two-step arithmetic handled correctly
3	A school library has 900 books. 60% are fiction. Of those, 25% are mystery novels. How many mystery novels are there?	Zero-shot	135	135	Correct	None	Nested percentage handled correctly; potential risk case
4	Maria sells 30 candles at $12 each. The material cost is $150. What is the profit?	Zero-shot	$210	$210	Correct	None	Simple revenue–cost calculation
5	A tank contains 200 litres of water. Each day, 15 litres evaporate and 40 litres are added. How much after 3 days?	Zero-shot	275 litres	275 litres	Correct	None	Multi-step daily update handled correctly

Few Shot Prompting

Q.No	Question	Prompt Type	Model Response	Actual Answer	Correct / Incorrect	Error Type	Remarks
6	A farmer has 120 eggs. He sells 40% and gives away 15 eggs. How many are left?	Few-shot	57 eggs	57 eggs	Correct	None	Examples helped sequence percentage then subtraction
7	A swimmer burns 50 calories per hour. Compare 3 hours vs 5 hours — how many more calories?	Few-shot	100 calories	100 calories	Correct	None	Example guided difference-based reasoning
8	Town A has 500 more people than B. B has 3 times C. If A = 2000, find C.	Few-shot	500 residents	500 residents	Correct	None	Strong structural similarity with example
9	In a class of 200 students, 30% play football. Of those, 40% play basketball. How many play both?	Few-shot	24 students	24 students	Correct	None	Avoided common percentage addition mistake
10	A cat eats 8 biscuits/day and a dog eats 3 times that. Over 7 days, how many total biscuits?	Few-shot	224 biscuits	224 biscuits	Correct	None	Correct multi-entity multiplication

Chain Of Thought Prompting

Q.No	Question	Prompt Type	Model Response	Actual Answer	Correct / Incorrect	Error Type	Remarks
11	A bookstore has 240 books. 25% are science, and 1/3 of those are children's books. How many children's science books?	Chain-of-Thought	20	20	Correct	None	Step-wise percentage prevents shortcut errors
12	Ann skates at 6 mph and Glenda at 8 mph in opposite directions for 3 hours. Distance apart?	Chain-of-Thought	42 miles	42 miles	Correct	None	CoT ensures both distances are computed
13	Tony increases his curl from 90 to 100 lbs. Military press = 2× curl, squat = 5× press. Find squat.	Chain-of-Thought	1000 lbs	1000 lbs	Correct	None	Prevented use of outdated value
14	84 scientists: half Europe, quarter Asia, rest USA. How many from USA?	Chain-of-Thought	21	21	Correct	None	Explicit subtraction avoids logical shortcut error
15	Apples $2, oranges $3, bananas $1. Buys 5, 4, 6 respectively. Pays $50. Change?	Chain-of-Thought	$22	$22	Correct	None	Multi-step calculation handled cleanly